MN - cursul examen 4

Capitol 7

$A \in \mathbb{R}^{n \times n}$

$Ax = \lambda x$ ecuație caracteristică

$x \neq 0$

$P_A(\lambda) = \det(A - \lambda I_n) = |A - \lambda I_n|$ polinom caracteristic

$\lambda(A) = \{\lambda_1, \dots, \lambda_n\}$ spectrul matricei A

$\rho(A) = \max_{i=1:n} \{|\lambda_i|\}$ raza spectrală

Proprietăți:

1) $\lambda(A - \mu I_n) = \{\lambda_1 - \mu, \lambda_2 - \mu, \dots, \lambda_n - \mu\}, \mu \in \mathbb{C}$

Dem: $Ax = \lambda x$

$Ax - \mu I_n x = \lambda x - \mu I_n x$

$(A - \mu I_n)x = (\lambda - \mu)x$ q.e.d.

2) $\lambda(A^{-1}) = \left\{\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}\right\}$

Dem: $Ax = \lambda x$

$x = \lambda A^{-1} x \mid : \lambda$

$A^{-1}x = \frac{1}{\lambda} x$

3) $\lambda(A^k) = \{\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k\}$

Dem: $Ax = \lambda x$

$A^2 x = \lambda A x = \lambda \cdot \lambda x = \lambda^2 x$

Inducție mat: $A^k x = \lambda^k x$

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4) $\lambda\left((A-\mu I_4)^{-1}\right) = \left\{\frac{1}{\lambda_1-\mu}, \frac{1}{\lambda_2-\mu}, \dots, \frac{1}{\lambda_4-\mu}\right\}$

$\mu \in \mathbb{C} \setminus \lambda(A)$

Dem: $(A-\mu I_4)x = (\lambda-\mu)x \mid \cdot (A-\mu I)^{-1}$

$x = (\lambda-\mu)(A-\mu I_4)^{-1}x \mid : \frac{1}{\lambda-\mu}$

$(A-\mu I)^{-1}x = \frac{1}{\lambda-\mu}x$

Metoda puterii directe (MPD)

Presupunem $|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_n|$

MPD calculează (λ_1, x_1)

Şirul $y^{(0)} \neq 0$

$y^{(1)} = Ay^{(0)}$ $\quad \lambda^{(1)}$

$y^{(2)} = Ay^{(1)} = A^2y^{(0)}$ $\quad \lambda^{(2)}$

$\vdots$

$\boxed{y^{(k)} = A \cdot y^{(k-1)} = A^k \cdot y^{(0)}}$ $\quad \boxed{\lambda^{(k)}}$

converge către x_1

$Ay^{(k)} = \lambda^{(k)}y^{(k)} \Rightarrow (y^{(k)})^t Ay^{(k)} = \lambda^{(k)}(y^{(k)})^t y^{(k)}$

$$\boxed{\lambda^{(k)} = \frac{(y^{(k)})^t Ay^{(k)}}{(y^{(k)})^t \cdot y^{(k)}}}$$ câtul Rayleigh

Algoritm MPD

D.i. : $A, y^{(0)}$, tol, maxiter

D.o. : λ_1, x_1

for $k = 1 : \text{maxiter}$

$z = A \cdot y^{(k-1)}$

$y^{(k)} = z / \text{norm}(z)$

$\lambda^{(k)} = (y^{(k)})' A y^{(k)}$

if $(\text{norm}(Ay^{(k)} - \lambda^{(k)} y^{(k)}) < \text{tol})$

return $(\lambda^{(k)}, y^{(k)})$

ex : 1) $A = \begin{bmatrix} -7 & 2 \\ 8 & -1 \end{bmatrix}$, $y^{(0)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\lambda_1 = -9$, $\lambda_2 = 1$

$y^{(1)} = \dfrac{Ay^{(0)}}{\|Ay^{(0)}\|} = \begin{bmatrix} -0,65 \\ 0,75 \end{bmatrix}$, $\lambda^{(1)} = (y^{(1)})' A y^{(1)} = -8,55$

$y^{(2)} = \dfrac{Ay^{(1)}}{\|Ay^{(1)}\|} = \begin{bmatrix} 0,71 \\ -0,71 \end{bmatrix}$, $\lambda^{(2)} = (y^{(2)})' A y^{(2)} = -9,04$

$\vdots$

2) $A = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$, $y^{(0)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\lambda_1 = 1$, $\lambda_2 = -1$

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$$y^{(1)} = \frac{Ay^{(0)}}{\|Ay^{(0)}\|} = \begin{bmatrix} 0,89 \\ -0,44 \end{bmatrix}, \quad \lambda^{(1)} = (y^{(1)})^T A y^{(1)} = 0,2$$

$$y^{(2)} = \frac{Ay^{(1)}}{\|Ay^{(1)}\|} = \begin{bmatrix} 0,707 \\ 0,707 \end{bmatrix}, \quad \lambda^{(2)} = (y^{(2)})^T A y^{(2)} = 0,5$$

$$y^{(3)} = \begin{bmatrix} 6,89 \\ -0,44 \end{bmatrix}, \quad \lambda^{(3)} = 0,2$$

$$y^{(4)} = \begin{bmatrix} 0,707 \\ 0,707 \end{bmatrix}, \quad \lambda^{(4)} = 0,5$$

Convergența șirului $y^{(k)}$, $k \geq 0$

$$y^{(k)} = \frac{A^k y^{(0)}}{\|A^k y^{(0)}\|}, \quad k \geq 0$$

Not. $\rho_k = \frac{1}{\|A_k \cdot y^{(0)}\|}$

Fie $y^{(0)} = \sum_{i=1}^{n} \alpha_i x_i$

$$y^{(k)} = \rho_k \cdot A^k \cdot \sum_{i=1}^{n} \alpha_i x_i = \rho_k \sum_{i=1}^{n} \alpha_i A^k x_i =$$

$$= \rho_k \sum_{i=1}^{n} \alpha_i \lambda_i^k x_i = \rho_k \alpha_1 \lambda_1^k x_1 + \rho_k \sum_{i=2}^{n} \alpha_i \lambda_i^k x_i =$$

$$= \rho_k \alpha_1 \lambda_1^k \left[x_1 + \sum_{i=2}^{n} \frac{\alpha_i}{\alpha_1} \left(\frac{\lambda_i}{\lambda_1} \right)^k x_i \right]$$

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Cea mai mică valoare proprie (λ_n)

$$|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_{n-1}| > |\lambda_n|$$

$$\lambda(A^{-1}) = \left\{ \frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_n} \right\}$$

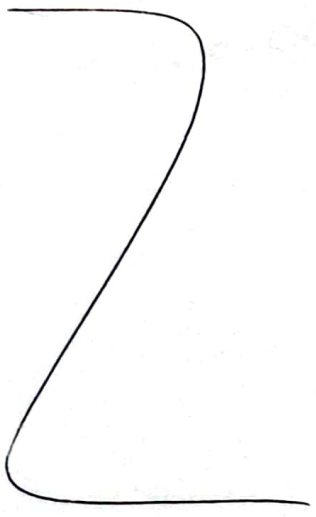
$$\left|\frac{1}{\lambda_n}\right| > \left|\frac{1}{\lambda_{n-1}}\right| \geq \dots \geq \left|\frac{1}{\lambda_2}\right| > \left|\frac{1}{\lambda_1}\right|$$

Algoritm:

D.i.: A, y_0, maxiter, tol

D.o.: (λ_n, x_n)

```
for k = 1 : maxiter
    z = A^{-1} y^{(k-1)}        % rezolvăm SEL  A z = y^{(k-1)}
    y^{(k)} = z / norm(z)
    λ^{(k)} = (y^{(k)})^T A^{-1} y^{(k)}
    if || A^{-1} y^{(k)} - λ^{(k)} y^{(k)} || < tol
        break
return 1/λ^{(k)}
```



Metoda puterii inverse

Algoritm:

D.I. : A, y_0, maxiter, tol, μ

D.O. : (λ, x)

for $k = 1$: maxiter

$z = (A - \mu I_n)^{-1} y^{(k-1)}$

% z este SEL $(A - \mu I) z = y^{(k-1)}$

$y^{(k)} = z / \text{norm}(z)$

$\lambda^{(k)} = (y^{(k)})^T A y^{(k)}$

if $\| Ay^{(k)} - \lambda^{(k)} y^{(k)} \| < \varepsilon$

return $(\lambda^{(k)}, y^{(k)})$

Localizarea valorilor proprii → Cercurile Gershgorin

→ vezi lab.

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Capitolul 8

Matrice Asemenea

$A, B \in \mathbb{R}^{n \times n}$ asemenea $\Leftrightarrow \exists\, T \in \mathbb{R}^{n \times n}$ nesingulară a.î.

$$\boxed{B = T^{-1}AT}$$

Proprietăți mat. asemenea:

1) $\boxed{\lambda(A) = \lambda(B)}$

Demi: $|B - \lambda I| = |T^{-1}AT - \lambda T^{-1}T| = |T^{-1}(A - \lambda I)T| =$

$= \det(T^{-1}) \cdot \det(A - \lambda I) \cdot \det(T) =$

$= \det(A - \lambda I)$

2) Fie (λ, x) pereche proprie pt. A
(λ, y) ——— " ——— B $\Rightarrow \boxed{x = Ty}$

Demi: $B \cdot (T^{-1}x) = T^{-1}AT \cdot T^{-1}x = T^{-1}Ax = T^{-1}\lambda x =$

$= \lambda T^{-1}x \Rightarrow$

$\Rightarrow B \cdot (T^{-1} \cdot x) = \lambda (T^{-1}x)$

$y = T^{-1}x \Rightarrow x = Ty$ q.e.d.

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Aducerea unei matrice la forma superior Hessenberg

O matrice $A \in \mathbb{R}^{n \times n}$ poate fi adusă la forma superior Hessenberg H_n folosind transformarea de asemănare ortogonală

$$\boxed{H_n = PAP^T}, \quad P \to \text{matrice ortogonală}$$

(A sim $\Rightarrow$ mat. ~~sup~~. tridiagonală / sup. Hes. + sim)

Fie $\boxed{P = P_{n-2} \cdot \ldots \cdot P_2 \cdot P_1}$, unde $P_1, P_2, \ldots, P_{n-2}$ sunt matrice elementare ~~Hausse~~ Householder

$$PAP^T = P_{n-2} \cdot \underbrace{P_2 P_1 A P_1^T P_2^T} \ldots P_{n-2}^T$$

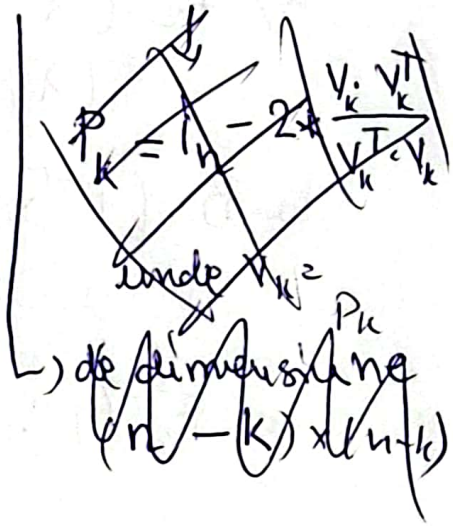
Fie $A_0 = A$

$A_1 = P_1 A_0 P_1^T$

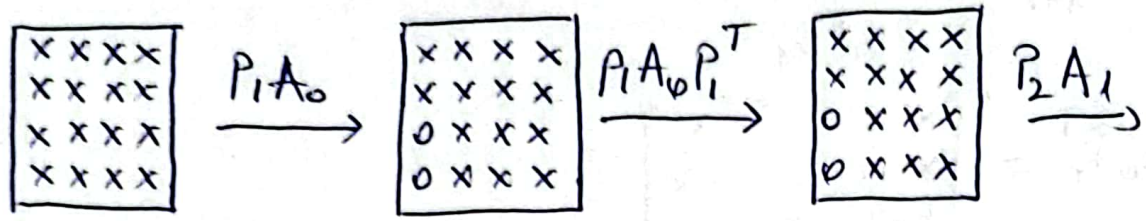
$A_2 = P_2 A_1 P_2^T$

$$\boxed{H_n = P_{n-2} A_{n-3} P_{n-2}^T}$$

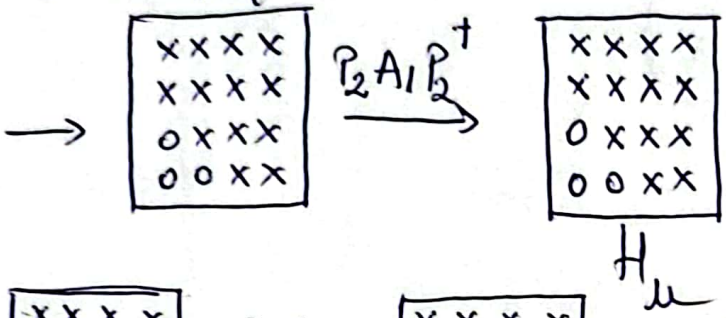
pt $n = 4$ $\longrightarrow$



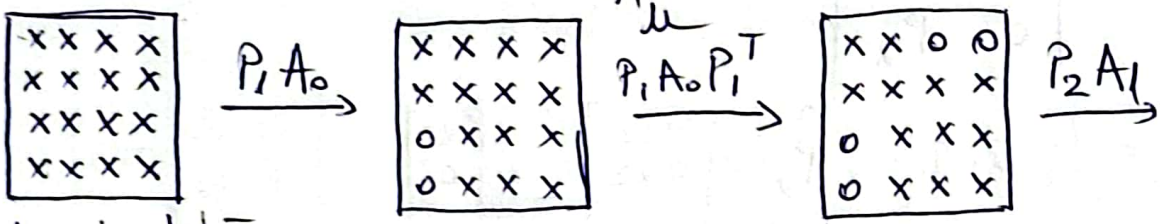
$n = 4$:



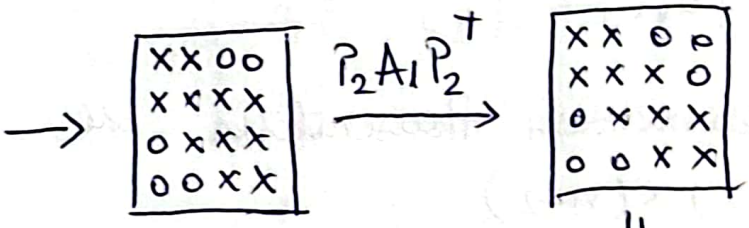
A_0 nesimetrică



H_u



A_0 simetrică



H_u

Pasul 1:

Fie $P_1 = \begin{bmatrix} 1 & 0 \\ 0 & \hat{P}_1 \end{bmatrix}$, $A_0 = A = \begin{bmatrix} x & \cdots & x \\ a_{21} & x & \cdots & x \\ a_{31} & x & \cdots & x \\ \vdots & & & \\ a_{n1} & x & \cdots & x \end{bmatrix}$

$\hat{P}_1 = ?$ a.î. $\hat{P}_1 \cdot \begin{bmatrix} a_{21} \\ a_{31} \\ \vdots \\ a_{n1} \end{bmatrix} = \begin{bmatrix} x \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$

$\hat{P}_1$ matricea elementară Householder de dimensiune $(n-1) \times (n-1)$

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$$\hat{P}_1 = I_{n-1} - \frac{2v_1 v_1^T}{2v_1^T v_1}$$

$$P_1 A_0 P_1^T = \begin{bmatrix} x & x & \cdots & x \\ x & x & \cdots & x \\ 0 & a_{32} & x & x \\ \vdots & \vdots & & \\ 0 & a_{n2} & x & x \end{bmatrix}$$

Pasul 2

$$P_2 = \begin{bmatrix} I_2 & 0 \\ 0 & \hat{P}_2 \end{bmatrix}$$

$\hat{P}_2 = ?$ a.î. $\hat{P}_2 \cdot \begin{bmatrix} a_{32} \\ a_{42} \\ \vdots \\ a_{n2} \end{bmatrix} = \begin{bmatrix} x \\ 0 \\ \vdots \\ 0 \end{bmatrix}$

$\hat{P}_2$ este matricea elementară Householder de dimensiune $(n-2) \times (n-2)$

$$\hat{P}_2 = I_{n-2} - 2 \cdot \frac{v_2 v_2^T}{v_2^T v_2}$$

$$A_2 = P_2 A_1 P_2^T$$

$\vdots$

Exemplu:

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

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Fie $P_1 = \begin{bmatrix} 1 & 0 \\ 0 & \hat{P}_1 \end{bmatrix}$

$\hat{P}_1 = ?$ a. î. $\hat{P}_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ 0 \end{bmatrix}$

$$\hat{P}_1 = I_2 - \frac{2 v_1 v_1^T}{v_1^T v_1}$$

$$v_1 = \begin{bmatrix} 1 + \nabla_1 \\ 1 \end{bmatrix} \qquad \nabla_1 = \operatorname{semn}(1) \cdot \left\| \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\| = \sqrt{2}$$

$$\|v_1\| = \sqrt{4 + 2\sqrt{2}}$$

$$\hat{P}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{2}{4 + 2\sqrt{2}} \begin{bmatrix} 1 + \sqrt{2} \\ 1 \end{bmatrix} \begin{bmatrix} 1 + \sqrt{2} & 1 \end{bmatrix}$$

$$\hat{P}_1 = \begin{bmatrix} -0{,}7 & -0{,}7 \\ -0{,}7 & 0{,}7 \end{bmatrix}$$

$$P_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0{,}7 & -0{,}7 \\ 0 & -0{,}7 & 0{,}7 \end{bmatrix}$$

$$H_u = P_1 A P_1^T = \begin{bmatrix} 0 & -2{,}12 & 0{,}7 \\ -1{,}41 & 3{,}5 & -0{,}5 \\ 0 & 1{,}5 & -0{,}5 \end{bmatrix}$$

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Forma Schur reală a unei matrice

(Algoritmul QR)

$\exists\, Q \in \mathbb{R}^{n\times n}$ ortogonală a.î. $S = Q^T A Q$, unde S s.n. forma Schur reală a mat. A.

(este matrice cvasi superior triunghiulară)

$$S = \begin{bmatrix} S_{11} & S_{12} & \cdots & S_{1n} \\ & S_{22} & \cdots & S_{2n} \\ & & \ddots & \vdots \\ 0 & & & S_{nn} \end{bmatrix}$$

Algoritmul QR fără deplasare pt. matrice nesimetrică

$A_0 = H_n$

$A_0 = Q_0 R_0$

$A_1 = R_0 Q_0 = Q_1 R_1$

$A_2 = R_1 Q_1 = Q_2 R_2$

$\vdots$

$A_k = R_{k-1} \cdot Q_{k-1}$

A_1 este ortogonal asemenea cu A_0

$$\left.\begin{aligned} A_1 &= R_0 Q_0 \\ A_0 &= Q_0 R_0 \Rightarrow R_0 = Q_0^T A_0 \end{aligned}\right| \Rightarrow A_1 = Q_0^T A_0 Q_0$$

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A_2 este ortogonal asemănea cu A_0.

$$A_2 = R_1 Q_1$$
$$A_1 = Q_1 R_1 \Rightarrow R_1 = Q_1^T A_1 \quad \Big| \Rightarrow$$
$$\Rightarrow A_2 = Q_1^T A_1 Q_1 = Q_1^T Q_0^T A_0 Q_0 Q_1 \Rightarrow$$
$$\Rightarrow A_2 = (Q_0 Q_1)^T \cdot A_0 \cdot Q_0 Q_1$$

....

Algoritm:

calculăm H_n

$A_0 = H_n$

for $k = 1 : maxiter$

$[Q_{k-1}\ R_{k-1}] = householder(A_{k-1})$

$A_k = R_{k-1} Q_{k-1}$

test convergență

Algoritmul QR cu deplasare simplă pt. matrice nesimetrică A

calculăm H_n

$A_0 = H_n$

for $k = 1 : maxiter$

$\mu_k = A_k(n, n)$

$[Q_k\ R_k] = householder(A_k - \mu_k I)$

$A_{k+1} = R_k Q_k + \mu_k I_n$

test convergență algoritm

Algoritmul QR fără deplasare pentru matrice simetrică A

calculare H_u

$A_0 = H_u$

for $k = 1 : maxiter$

$[Q_k \; R_k] = householder(A_k)$

$A_{k+1} = R_k \cdot Q_k$

test convergență algoritm

Algoritmul QR cu deplasare simplă pentru matrice simetrică A

calculare H_u

$A_0 = H_u$

for $k = 1 : maxiter$

calculare μ_k

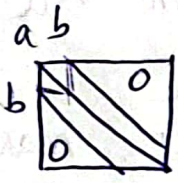
$[Q_k \; R_k] = householder(A_k - \mu_k I_n)$

$A_{k+1} = R_k Q_k + \mu_k I_n$

test convergență algoritm

→ Calculare μ_k :

Fie $E^{(k)} = \begin{bmatrix} a_{n-1}^{(k)} & b_{n-1}^{(k)} \\ b_{n-1}^{(k)} & a_n^{(k)} \end{bmatrix}$



$$|E^{(k)} - \lambda I_2| = \begin{vmatrix} a_{n-1}^{(k)} - \lambda & b_{n-1}^{(k)} \\ b_{n-1}^{(k)} & a_n^{(k)} - \lambda \end{vmatrix} =$$

$$= (a_{n-1}^{(k)} - \lambda)(a_n^{(k)} - \lambda) - (b_{n-1}^{(k)})^2 =$$

$$= a_{n-1}^{(k)} \cdot a_n^{(k)} - \lambda \cdot a_{n-1}^{(k)} - \lambda \cdot a_n^{(k)} + \lambda^2 - (b_{n-1}^{(k)})^2 =$$

$$= \lambda^2 - \lambda (a_{n-1}^{(k)} + a_n^{(k)}) + a_{n-1}^{(k)} \cdot a_n^{(k)} - (b_{n-1}^{(k)})^2$$

$\Rightarrow$ rădăcinile λ_1 și λ_2

μ_k va fi egal cu valoarea proprie (λ_1 sau λ_2) cea mai apropiată de elementul $a_n^{(k)}$.

Descompunerea valorilor singulare (SVD)

$A = U \Sigma V^T$, $A \in \mathbb{R}^{m \times n}$

$U = [u_1 \; u_2 \; \dots \; u_m] \in \mathbb{R}^{m \times m}$ matrice ortogonală

$u_1, u_2, \dots, u_m$ s.n. vectori singulari stânga ai matricei A

$$V^T = \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{bmatrix} \in \mathbb{R}^{n \times n}$$ matrice ortogonală

$v_1, v_2, \dots, v_n$ s.n. vectori singulari dreapta ai matricei A.

$\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_p) \in \mathbb{R}^{m \times n}$, $p = \min(m, n)$

$\{\sigma_k\}$ → valorile singulare ale matricei A

$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$

$\sigma_{r+1} = \sigma_{r+2} = \dots = \sigma_p = 0$

Calculare Σ !

$$\boxed{\sigma_i = \sqrt{\lambda_i}, \text{ unde } \lambda_i \in \lambda(A^T \cdot A)}$$

Dem: $A = U \Sigma V^T$

$AV = U\Sigma$

$[Av_1 \; Av_2 \dots Av_n] = [\sigma_1 u_1 \; \sigma_2 u_2 \dots \sigma_n u_n]$

$Av_i = \sigma_i u_i, \; i = 1 : n \quad (1)$

$A = U\Sigma V^T \Rightarrow A^T = V\Sigma^T U^T \Rightarrow A^T = V\Sigma U^T$

$A^T U = V\Sigma$

$Au_i = \sigma_i v_i, \; i = 1 : m \quad (2)$

Din (1) $\Rightarrow$ $Av_i = \sigma_i u_i$

$A^T A v_i = \sigma_i \cdot A^T u_i \overset{(2)}{\Rightarrow}$

$\Rightarrow \boxed{A^T A v_i = \sigma_i^2 v_i} \Rightarrow$

~~$\Rightarrow A^T A v_i \in \mathbb{R}$~~ $\rightarrow (3)$

$\Rightarrow \boxed{\lambda_i = \sigma_i^2}$

Din (2) $\Rightarrow$ $A^T u_i = \sigma_i v_i$

$AA^T u_i = \sigma_i A v_i$ $\overset{(1)}{=}$

$\Rightarrow AA^T u_i = \sigma_i^2 u_i \Rightarrow \boxed{\lambda_i = \sigma_i^2}$

Calculare V

Din (3) $\Rightarrow A^T A v_i = \sigma_i^2 v_i$, $i = 1:n$

v_i sunt vectorii proprii ai matricei $A^T A$

$V = [v_1 \; v_2 \; \dots \; v_n]$

Calculare U

Din (1) $\Rightarrow A v_i = \sigma_i u_i$

$$\boxed{u_i = \frac{1}{\sigma_i} A v_i, \quad i = 1:r}$$

$U = [u_1 \; u_2 \; \dots \; u_m] =$

$= [u_1 \; u_2 \; \dots \; u_r \; u_{r+1} \; \dots \; u_m]$

$\downarrow$

se calculează folosind alg. Gram–Schmidt

ex: vezi lab (sau table)

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Proprietăți SVD : $A \in \mathbb{R}^{m \times n}$

1) $\boxed{\|A\|_2 = \sigma_1}$

Dem : $\|A\|_2 = \|U \Sigma V^T\|_2 = \|\Sigma V^T\|_2 = \|\Sigma\|_2 =$ (U ortog., V^T ortog.)

$= \sqrt{\rho(\Sigma^T \Sigma)} = \sqrt{\max\{\sigma_1^2, \sigma_2^2, \dots\}} = \sigma_1$

2) $\|A\|_F = \sqrt{\sigma_1^2 + \sigma_2^2 + \dots + \sigma_n^2}$

$\|A\|_F = \|U \Sigma V^T\|_F = \|\Sigma\|_F = \sqrt{\sum_{i=1}^{n} \sigma_i^2}$.

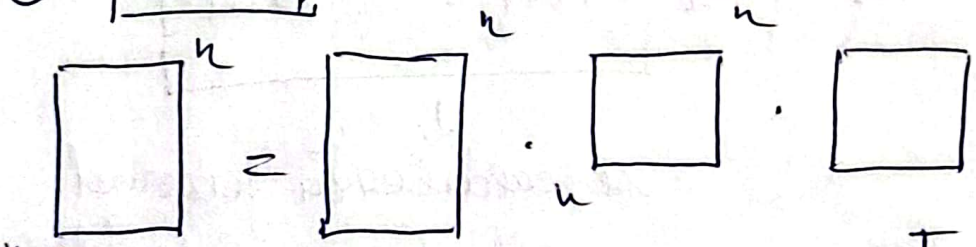
3) $\|A^{-1}\|_2 = \frac{1}{\sigma_n}$

4) $\text{cond}(A) = \|A\|_2 \cdot \|A^{-1}\|_2 = \frac{\sigma_1}{\sigma_n}$

5) $\text{rang}(A) =$ nr. de valori singulare > 0

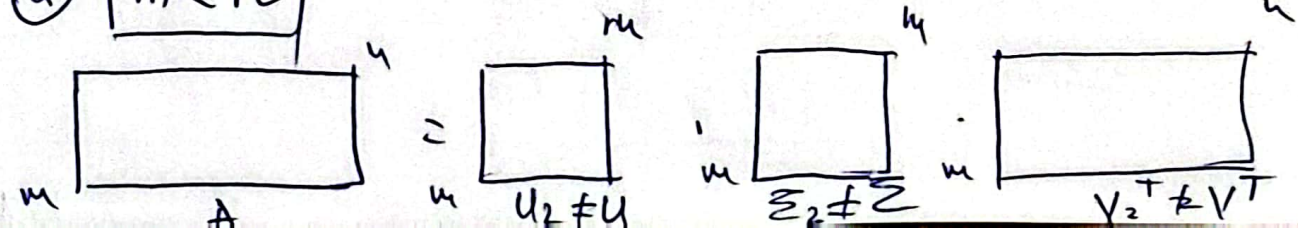
Teoreme : $A \in \mathbb{R}^{m \times n}$, $A = U_1 \Sigma_1 V_1^T$

① $\boxed{m > n}$



(m×n) (m×n) (n×n) (n×n)

② $\boxed{m < n}$



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Teorema 3

Fie $A \in \mathbb{R}^{m \times n}$

$$A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_n u_n v_n^T$$

$m \cdot \| \cdot \|$

Compresia imaginilor → vezi table ...

$\|A\|_\infty$ = max suma modulelor pe linii

$\|A\|_2 = \sqrt{\rho(A^T \cdot A)}$

$\rho(A)$ = max val pp.

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Interpolare

Fie funcția $f: [x_0, x_n] \to \mathbb{R}$ cunoscută doar în $x_0, x_1, \dots, x_n$ prin valorile $f(x_0), f(x_1), \dots, f(x_n)$

$x_0 < x_1 < \dots < x_n$

$\{x_0, \dots, x_n\}$ = suportul interpolării

Fie <u>polinomul de interpolare</u> de gradul n

$P_n(x) = a_0 u_0(x) + a_1 u_1(x) + \dots + a_n u_n(x)$,

unde $\{u_0(x), u_1(x), \dots, u_n(x)\}$ <u>baza de interpolare</u>

$f(x) = ?$, $x \in [x_0, x_n] \setminus \{$suportul interpolării$\}$

$f(x) \cong P_n(x)$

→pt. a calcula coeficienții $a_0, a_1, \dots, a_n$ se folosesc <u>condițiile de interpolare</u>:

$$\boxed{f(x_i) = P_n(x_i),\ i = \overline{0,n}}$$

Interpolare polinomială

Se folosește baza de interpolare $\{1, x, \dots, x^n\}$.

Pt. a calcula coef. $a_0, a_1, \dots, a_n$ se fol. cond. de interp.: $f(x_i) = P_n(x_i)$, $i = 0:n$.

$$P_n(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$$

$$\begin{cases} a_0 + a_1 x_0 + a_2 x_0^2 + \dots + a_n x_0^n = f(x_0) \\ a_0 + a_1 x_1 + a_2 x_1^2 + \dots + a_n x_1^n = f(x_1) \\ \vdots \\ a_0 + a_1 x_n + a_2 x_n^2 + \dots + a_n x_n^n = f(x_n) \end{cases}$$

Sistem de $n+1$ ecuații liniare:

$$\begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & & & & \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix} \cdot \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} f(x_0) \\ f(x_1) \\ \vdots \\ f(x_n) \end{bmatrix}$$

↳ matricea Vandermonde

(comanda Octave : vander)

↓

mat. rău condiționată

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Interpolare Lagrange

Se foloseste baza de interpolare $\{L_0(x), L_1(x), \dots, L_n(x)\}$

$$L_0(x) = \frac{(x-x_1)(x-x_2)\dots(x-x_n)}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)}$$

$$\vdots$$

$$L_n(x) = \frac{(x-x_0)(x-x_1)\dots(x-x_{n-1})}{(x_n-x_0)(x_n-x_2)\dots(x_n-x_{n-1})}$$

$$\boxed{L_k(x) = \prod_{\substack{i=0 \\ i\neq k}}^{n} \frac{x-x_i}{x_k-x_i}} \rightarrow \text{multiplicatorul Lagrange (pol. de gr } n)$$

$$P_n(x) = \sum_{k=0}^{n} \underbrace{a_k}_{f(x_k)} \cdot \underbrace{u_k(x)}_{L_k(x)} \Rightarrow \boxed{P_n(x) = \sum_{k=0}^{n} f(x_k) \cdot \prod_{\substack{i=0 \\ i\neq k}}^{n} \frac{x-x_i}{x_k-x_i}}$$

$P_n(x_i) = f(x_i)$

$L_k(x_i) = \begin{cases} 1, & k=i \\ 0, & k\neq i \end{cases}$

Ex:

x	1	2	3	4
$f(x)$	1	2	0	1

Polinomul de interpolare Lagrange $P_3(x) = ?$

$x_0 = 1,\ x_1 = 2,\ x_2 = 3,\ x_3 = 4$

$$P_3(x) = a_0L_0(x) + a_1L_1(x) + a_2L_2(x) + a_3L_3(x)$$

$a_0 = f(x_0) = 1$
$a_1 = f(x_1) = 2$
$a_2 = f(x_2) = 0$
$a_3 = f(x_3) = 1$

$$L_0(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)}$$

$$\vdots$$

$$L_3(x) = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}$$

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Interpolare Newton

Diferențe divizate:

F_0 (diferențe divizate de ordinul 0) $\quad$ F_1 (dif. div. de ord 1) $\quad$ F_2 (dif. div. de ord 2)

$$F_0[x_0] = f(x_0) \longrightarrow F_1[x_0, x_1] = \frac{F_0[x_0] - F_0[x_1]}{x_0 - x_1}$$

$$F_0[x_1] = f(x_1)$$

$$F_0[x_2] = f(x_2) \longrightarrow F_1[x_1, x_2] = \frac{F_0[x_1] - F_0[x_2]}{x_1 - x_2}$$

$$F_0[x_n] = f(x_n)$$

$$F_2[x_0, x_1, x_2] = \frac{F_1[x_0, x_1] - F_1[x_1, x_2]}{x_0 - x_2}$$

$$\Rightarrow \boxed{F_n[x_0, x_1, \dots, x_n] = \frac{F_{n-1}[x_0, x_1, \dots, x_{n-1}] - F_{n-1}[x_1, x_2, \dots, x_n]}{x_0 - x_n}}$$

$$F_1[x, x_0] = \frac{F_0[x] - F_0[x_0]}{x - x_0} = \frac{f(x) - f(x_0)}{x - x_0}$$

$$(x - x_0) F_1[x, x_0] = f(x) - f(x_0)$$

$$(x - x_1) F_2[x, x_0, x_1] = F_1[x, x_0] - F_1[x_0, x_1] \;/\cdot (x - x_0)$$

$$(x - x_2) F_3[x, x_0, x_1, x_2] = F_2[x, x_0, x_1] - F_2[x_0, x_1, x_2] \;/\cdot (x - x_0)(x - x_1)$$

$$\vdots$$

$$(x - x_n) F_{n+1}[x, x_0, \dots, x_n] = F_n[x, x_0, \dots, x_{n-1}] - F_n[x_0, \dots, x_n] \;/ (x - x_0)(x - x_1)\cdots(x - x_n)$$

$$f(x) = f(x_0) + F_1[x_0, x_1](x - x_0) + F_2[x_0, x_1, x_2](x - x_0)(x - x_1) + \dots + F_n[x_0, x_1, \dots, x_n](x - x_0)(x - x_1)\cdots(x - x_{n-1}) + F_{n+1}[x, x_0, \dots, x_n](x - x_0)(x - x_1)\cdots(x - x_n)$$

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$f(x) = P_n(x) + E(x)$

$P_n(x)$ ↓ polinom de interpolare

$E(x)$ ↓ eroarea interpolării

$\{u_0(x), u_1(x), \dots\}$

Baza de interpolare: $\{1, (x-x_0), (x-x_0)(x-x_1), \dots, (x-x_0)(x-x_1)\cdots(x-x_{n-1})\}$

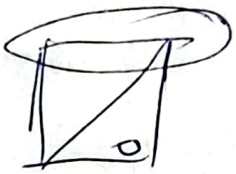
$a_0 = f(x_0) = F_0[x_0]$

$a_1 = F_1[x_0, x_1]$

$a_2 = F_2[x_0, x_1, x_2]$

$\vdots$

$a_n = F_n[x_0, x_1, \dots, x_n]$



Ex:

x	1	2	3	4
$f(x)$	1	2	0	1

$x_0 = 1$
$x_1 = 2$
$x_2 = 3$
$x_3 = 4$

x	F_0		F_1		F_2		F_3
$x_0=1$	$F_0[x_0] = f(x_0) = 1$	→	$F_1[x_0, x_1] = \frac{F_0[x_0] - F_0[x_1]}{x_0 - x_1} = \frac{1-2}{1-2} = 1$				
$x_1=2$	$F_0[x_1] = f(x_1) = 2$	↗↘	$F_1[x_1, x_2] = \frac{F_0[x_1] - F_0[x_2]}{x_1 - x_2} = \frac{2-0}{2-3} = -2$	→	$F_2[x_0, x_1, x_2] = \frac{F_1[x_0, x_1] - F_1[x_1, x_2]}{x_0 - x_2} = \frac{1-(-2)}{1-3} = -\frac{3}{2}$		
$x_2=3$	$F_0[x_2] = f(x_2) = 0$	↗↘	$F_1[x_2, x_3] = \frac{F_0[x_2] - F_0[x_3]}{x_2 - x_3} = \frac{0-1}{3-4} = 1$	→	$F_2[x_1, x_2, x_3] = \frac{-2-1}{2-4} = \frac{3}{2}$	↗↘	$F_3[x_0, x_1, x_2, x_3] = \frac{-3/2 - 3/2}{1-4} = 1$
$x_3=4$	$F_0[x_3] = f(x_3) = 1$	↗					

$P_3(x) = a_0 u_0(x) + a_1 u_1(x) + a_2 u_2(x) + a_3 u_3(x)$

$P_3(x) = F_0[x_0] + (x-x_0) F_1[x_0, x_1] + (x-x_0)(x-x_1) F_2[x_0, x_1, x_2] + (x-x_0)(x-x_1)(x-x_2) F_3[x_0, x_1, x_2, x_3]$

Interpolare liniară

Un spline $S: [x_0, x_n] \to \mathbb{R}$ este o funcție

$$S(x) = \begin{cases} S_0(x), & x \in [x_0, x_1] \\ S_1(x), & x \in [x_1, x_2] \\ \vdots \\ S_{n-1}(x), & x \in [x_{n-1}, x_n] \end{cases}, \text{ unde } S_i: [x_i, x_{i+1}] \to \mathbb{R}$$

fie $\boxed{S_i(x) = a_i x + b_i}$, $i = \overline{0, n-1}$

$\Downarrow$

S s.n. spline liniar

Calculare coeficienți a_i, b_i, $i = \overline{0, n-1}$ (2n coef.)

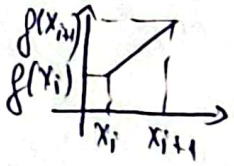
Metoda 1:

Condiții interpolare: $\begin{cases} S_i(x_i) = f(x_i), & i = \overline{0, n-1} \\ S_{n-1}(x_n) = f(x_n) \end{cases}$

Condiții de racordare: $S_i(x_{i+1}) = S_{i+1}(x_{i+1})$, $i = \overline{0, n-2}$

$\Rightarrow$ sistem liniar de $2n$ ecuații liniare

Metoda 2:



- fie ec. dreptei care trece prin $(x_i, f(x_i))$, $(x_{i+1}, f(x_{i+1}))$

$$\frac{y - f(x_i)}{f(x_{i+1}) - f(x_i)} = \frac{x - x_i}{x_{i+1} - x_i}$$

$$y = f(x_i) + \frac{(x - x_i)(f(x_{i+1}) - f(x_i))}{x_{i+1} - x_i}$$

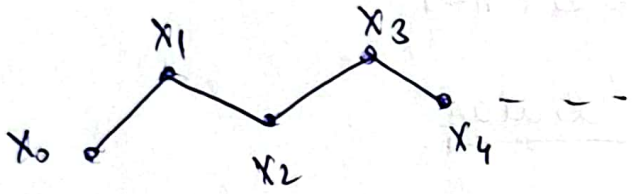
adică la ac. numitor, desfac parantezele

$$\Rightarrow y = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} x + \frac{f(x_i) x_{i+1} - f(x_{i+1}) x_i}{x_{i+1} - x_i}$$

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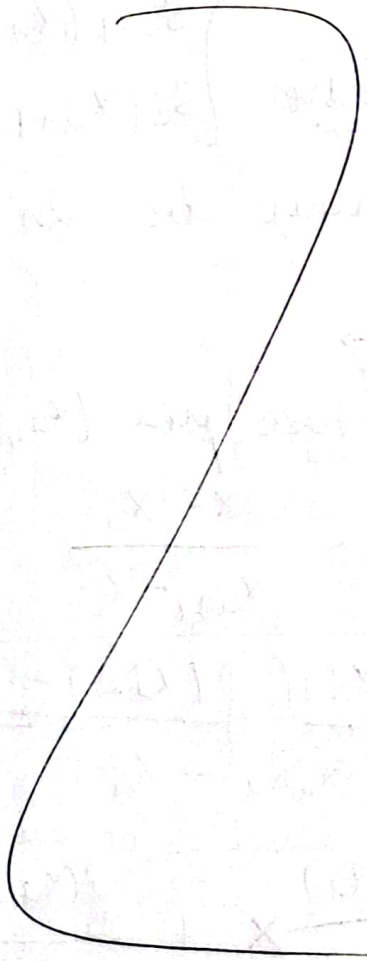
$$a_i = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}, \quad i = 0 : n-1$$

$$b_i = \frac{x_{i+1} f(x_i) - x_i f(x_{i+1})}{x_{i+1} - x_i}, \quad i = 0 : n-1$$



function val = luter_etu(x, y, c)

de ce
u-a
scris :



Polinoamele Bernstein de gradul n

$$B_{i,n}(t) = C_n^i (1-t)^{n-i} \cdot t^i, \quad i = 0:n$$

parametrul $t \in [0,1]$

$\underline{n=1}$: (Pol. Bernstein de gr I)

$B_{0,1}(t) = C_1^0 (1-t)^1 t^0 = 1-t$

$B_{1,1}(t) = C_1^1 (1-t)^0 t^1 = t$

$\underline{n=2}$: (Pol. Bernstein de gr 2)

$B_{0,2}(t) = (1-t)^2$

$B_{1,2}(t) = 2(1-t)t$

$B_{2,2}(t) = t^2$

$\underline{n=3}$: (Pol. Bernstein de gr 3)

$B_{0,3}(t) = (1-t)^3$

$B_{1,3}(t) = 3(1-t)^2 t$

$B_{2,3}(t) = 3(1-t)t^2$

$B_{3,3}(t) = t^3$

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Interpolare cubică de clasă C^1

Fie spline-ul cubic $S: [x_0, x_n] \to \mathbb{R}$

$$S(x) = \begin{cases} S_0(x), & x \in [x_0, x_1] \\ S_1(x), & x \in [x_1, x_2] \\ \vdots \\ S_{n-1}(x), & x \in [x_{n-1}, x_n] \end{cases}$$

unde $S_i(x) = a_i + (x - x_i) b_i + (x - x_i)^2 c_i + (x - x_i)^3 d_i$

$S_i : [x_i, x_{i+1}] \to \mathbb{R}$

$a_i, b_i, c_i, d_i = ?$, $i = 0 : n-1$ (4n necunoscute)

Condiții de interpolare de tip Hermite:

$$\begin{cases} S_i(x_i) = f(x_i), & i = 0 : n-1 \\ S_{n-1}(x_n) = f(x_n) \\ S_i'(x_i) = f'(x_i), & i = 0 : n-1 \\ S_{n-1}'(x_n) = f'(x_n) \end{cases}$$

Condiții de racordare:

$$\begin{cases} S_i(x_{i+1}) = S_{i+1}(x_{i+1}), & i = 0 : n-2 \\ S_i'(x_{i+1}) = S_{i+1}'(x_{i+1}), & i = 0 : n-2 \end{cases}$$

Spline-ul cubic de clasă C^1 în formă parametrică

$$\boxed{h_i = x_{i+1} - x_i}, \quad \boxed{t = \frac{x - x_i}{h_i}}$$

$$S_i(t) = a_i + b_i \cdot t \cdot h_i + c_i \cdot t^2 h_i^2 + d_i t^3 h^3, \quad i = 0 : n-1$$

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Spline-ul cubic de cls C1 în forma parametrică și folosind polinoamele Bernstein de gr 3 !

$$S_i(t) = a_i'(1-t)^3 + b_i' 3(1-t)^2 t + c_i' 3(1-t)t^2 + d_i' t^3, \quad i = 0:n-1$$

$$h_i = x_{i+1} - x_i$$
$$a_i' = f(x_i)$$
$$b_i' = f(x_i) + \frac{h_i}{3} f'(x_i)$$
$$c_i' = f(x_{i+1}) - \frac{h_i}{3} f'(x_{i+1})$$
$$d_i' = f(x_{i+1})$$

$t \to x$

Ex:

x	1	2	4
$f(x)$	3	4	6
$f'(x)$	0	2	5

$x_0 = 1$
$x_1 = 2$
$x_2 = 4$

$$S: [1,4] \to \mathbb{R}, \quad S(x) = \begin{cases} S_0(x), & x \in [1,2] \\ S_1(x), & x \in [2,4] \end{cases}$$

Metoda 1: $S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$

$S_i'(x) = b_i + 2c_i(x - x_i) + 3d_i(x - x_i)^2$

cond. de interp. de tip Hermite: $S_0(1) = f(1), \; i' = 0:1$
$S_1(2) = f(2)$
$S_1(4) = f(4)$
$S_0'(1) = f'(1)$
$S_1'(2) = f'(2)$
$S_1'(4) = f'(4)$

cond de racordare: $S_0(2) = S_1(2)$
$S_0'(2) = S_1'(2)$

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Metoda 2

$S_i(t) = a_i'(1-t)^3 + b_i' 3(1-t)^2 t + c_i' 3(1-t)t^2 + d_i' t^3$, $i = 0:1$

$a_0' = f(x_0) = 3$

$a_1' = f(x_1) = 4$

$h_0 = 2-1 = 1$

$h_1 = x_2 - x_1 = 4-2 = 2$

$b_0' = f(x_0) + \frac{h_0}{3} f'(x_0) = 3 + \frac{1}{3} \cdot 0 = 3$

$b_1' = f(x_1) + \frac{h_1}{3} f'(x_1) = 4 + \frac{2}{3} \cdot 2 = \frac{16}{3}$

$c_0' = f(x_1) - \frac{h_0}{3} f'(x_1) = 4 - \frac{1}{3} \cdot 2 = \frac{10}{3}$

$c_1' = f(x_2) - \frac{h_1}{3} f'(x_2) = 6 - \frac{2}{3} \cdot 3 = \frac{9}{3}$

$d_0' = f(x_1) = 4$

$d_1' = f(x_2) = 6$

$S_0(t) = a_0'(1-t)^3 + b_0' \dots$

sch. var: $t = \frac{x - x_0}{h_0} = x - 1$

$S_0(x) = 3(1 - x + 1)^3 + 3 \cdot 3(2-x)^2(x-1) + \frac{10}{3} \cdot 3 \cdot (2-x)(x-1)^2 + 4(x-1)$

$S_1(t) = \dots$

sch. var: $t = \frac{x - x_1}{h_1} = \frac{x-4}{2}$

$S_1(x) = \dots$

Interpolare cubică de clasă C2

Fie spline-ul cubic $S: [x_0, x_n] \to \mathbb{R}$

$$S(x) = \begin{cases} S_0(x), & x \in [x_0, x_1] \\ \vdots \\ S_{n-1}(x), & x \in [x_{n-1}, x_n] \end{cases}$$

unde $S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$, $i = 0 : n-1$

$a_i, b_i, c_i, d_i = ?$

Condiții de interpolare:

$$S_i(x_i) = f(x_i), \quad i = 0 : n-1$$
$$S_{n-1}(x_n) = f(x_n)$$

Condiții de racordare:

$$S_i(x_{i+1}) = S_{i+1}(x_{i+1}), \quad i = 0 : n-2$$
$$S_i'(x_{i+1}) = S_{i+1}'(x_{i+1}), \quad \text{---''---}$$
$$S_i''(x_{i+1}) = S_{i+1}''(x_{i+1}), \quad \text{---''---}$$

Spline cubic natural de clasă C2:

$$S_0''(x_0) = 0$$
$$S_{n-1}''(x_n) = 0$$

Spline cubic tensionat de clasă C2:

$$S_0'(x_0) = f'(x_0)$$
$$S_{n-1}'(x_n) = f'(x_n)$$

ex: doar nelocuiești (vezi tema 2 :))

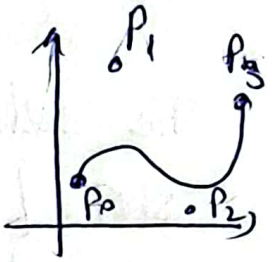
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Curbe Bezier

Fie pct. $P_0(x_0, y_0), P_1(x_1, y_1), \dots, P_n(x_n, y_n)$

$P_0, P_n \rightarrow$ puncte de interpolare

$P_1, d=\overline{1, n-1} \rightarrow$ puncte de control



$$B_n(t) = \sum_{i=0}^{n} P_i B_{i,n}(t), \quad t \in [0,1]$$

→ curba Bézier de ordinul n (pol. Bernstein de ord n)

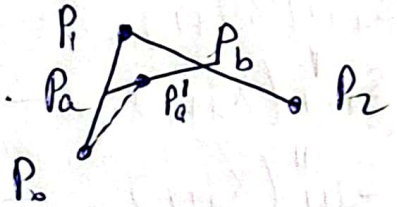
Curba Bezier dreaptă: $n=1$ (P_0, P_1)

$$B_1(t) = P_0 B_{0,1}(t) + P_1 B_{1,1}(t) = (1-t)P_0 + t \cdot P_1, \quad t \in [0,1]$$

$P_0 \longrightarrow A$

Curba Bezier pătratică: $n=2$ (P_0, P_1, P_2)

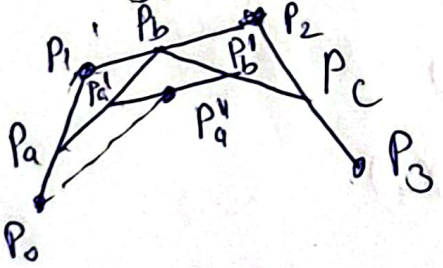
$$B_2(t) = P_0 B_{0,2}(t) + P_1 B_{1,2}(t) + P_2 B_{2,2}(t) = (1-t)^2 P_0 + 2t(1-t)P_1 + t^2 P_2, \quad t \in [0,1]$$



$$\begin{cases} P_0 \rightarrow P_a = (1-t)P_0 + tP_1 \\ P_1 \rightarrow P_b = (1-t)P_1 + tP_2 \\ P_2 \end{cases} \rightarrow P'_a = B_2(t) = (1-t)P_a + tP_b$$

Curba Bezier cubică: $n=3$ (P_0, P_1, P_2, P_3)

$$B_3(t) = (1-t)^3 P_0 + 3t(1-t)^2 P_1 + 3t^2(1-t)P_2 + t^3 P_3$$



$$\begin{cases} P_0 \rightarrow P_a = (1-t)P_0 + tP_1 \rightarrow P'_a = (1-t)P_a + tP_b \\ P_1 \rightarrow P_b = (1-t)P_1 + tP_2 \rightarrow P'_b = (1-t)P_b + tP_c \\ P_2 \rightarrow P_c = (1-t)P_2 + tP_3 \end{cases} \rightarrow P''_a = B_3(t) = (1-t)P'_a + tP'_b$$

Algoritmul de Casteljau

I.: $P_0(x_0, y_0), \dots, P_n(x_n, y_n)$, $(t \in [0,1])$

```
for j = 2 : n
    for i = 1 : n - j + 1
        P_i^(j) = (1 - t) P_i^(j-1) + t P_(i+1)^(j-1)
return P_1^(n)
```



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capitolul 10

Derivare numerică

$f'(x_0) = ?$

$$f(x) \simeq P_n(x) = \sum_{k=0}^{n} f(x_k) \cdot L_k(x) = \sum_{k=0}^{n} f(x_k) \cdot \prod_{\substack{i=0 \\ i \neq k}}^{n} \frac{x - x_i}{x_k - x_i}$$

n = 1 (x_0, x_1)

$$P_1(x) = f(x_0) L_0(x) + f(x_1) L_1(x)$$

$$f'(x_0) \simeq P_1'(x_0)$$

$$P_1'(x) = f(x_0) L_0'(x) + f(x_1) L_1'(x)$$

$$L_0'(x) = \frac{1}{x_0 - x_1} \; ; \; L_1'(x) = \frac{1}{x_1 - x_0}$$

Fie $x_0 = x$, $x_1 = x_0 + h$

$$f'(x_0) \simeq P_1'(x_0) = f(x_0) \cdot \frac{1}{x_0 - x_0 - h} + f(x_0 + h) \cdot \frac{1}{x_0 + h - x_0}$$

$$\boxed{f'(x_0) \simeq \frac{1}{h} [f(x_0 + h) - f(x_0)]}$$

n = 2 (x_0, x_1, x_2)

$$P_2(x) = f(x_0) L_0(x) + f(x_1) L_1(x) + f(x_2) L_2(x)$$

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} \; ; \; L_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}$$

$$L_2(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$

$$P_2'(x) = f(x_0) L_0'(x) + f(x_1) L_1'(x) + f(x_2) L_2'(x)$$

$$L_0'(x) = \frac{2x - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)}$$

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$f'(x) \simeq P_2'(x)$

Fie $x_0 = x$, $x_1 = x_0 + h$, $x_2 = x_0 + 2h$

$$f'(x_0) \simeq f(x_0) \cdot \frac{2x_0 - x_0 - h - x_0 - 2h}{(x_0 - x_0 - h)(x_0 - x_0 - 2h)} +$$

$$+ f(x_0+h) \cdot \frac{2x_0 - x_0 - x_0 - 2h}{(x_0 + h - x_0)(x_0 + h - x_0 - 2h)} +$$

$$+ f(x_0+2h) \cdot \frac{2x_0 - x_0 - x_0 - h}{(x_0 + 2h - x_0)(x_0 + 2h - x_0 - h}$$

$$\Rightarrow \boxed{f'(x_0) \simeq \frac{1}{h}\left[-\frac{3}{2}f(x_0) + 2f(x_0+h) - \frac{1}{2}f(x_0+2h)\right]}$$

Dacă $x = x_1$

$f'(x_1) \simeq P_2'(x_1)$

$x_0 \to x_0 - h$

$x_1 \to x_0$

$x_2 \to x_0 + h$

$$\Rightarrow \ldots \Rightarrow \boxed{f'(x_0) \simeq \frac{1}{2h}\left[f(x_0+h) - f(x_0-h)\right]}$$

$f''(x_0) = ?$

$$f(x) = f(x_0) + (x - x_0) f'(x_0) + \frac{(x-x_0)^2}{2} f''(x_0) + \ldots$$

(serie Taylor)

$x \to x_0 - h$

$$f(x_0 - h) \simeq f(x_0) + (-h) f'(x_0) + \frac{h^2}{2} f''(x_0) \quad (1)$$

$x \to x_0 + h$

$$f(x_0 + h) \simeq f(x_0) + h \cdot f'(x_0) + \frac{h^2}{2} f''(x_0) \quad (2)$$

$(1) + (2) \Rightarrow \ldots \Rightarrow f''(x_0) \simeq \frac{1}{h^2}\left[f(x_0 - h) - 2f(x_0) + f(x_0 + h)\right]$

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Integrare numerică – Metodele Newton – Cotes

$f:[a,b]\to\mathbb{R}$ continuă

$\int_a^b f(x)\,dx = ?$

Fie $x_k = a + kh,\ k=\overline{0,n}$, $h = \frac{b-a}{n}$

$$\int_a^b f(x)\,dx \approx \sum_{k=0}^{n} q_k f(x_k) \quad (1)$$

? tis. ofeal

$$f(x) \approx P_n(x) = \sum_{k=0}^{n} f(x_k) L_k(x) = \sum_{k=0}^{n} f(x_k) \cdot \prod_{\substack{i=0 \\ i\neq k}}^{n} \frac{x-x_i}{x_k-x_i}$$

$$\int_a^b f(x)\,dx \approx \int_a^b P_n(x)\,dx = \sum_{k=0}^{n} f(x_k) \cdot \int_a^b L_k(x)\,dx$$

Din (1) și (2) $\Rightarrow$ $q_k = \int_a^b L_k(x)\,dx$

Met. tr. → vezi lab (2 pct.)

Met simplă Simpson (3 pct.)

$n=2$:

$x_0 = a$, $x_1 = a+h = \frac{a+b}{2}$, $x_2 = b$, $h = \frac{b-a}{2}$

$f(x) = f(x_1) + (x-x_1) f'(x_1) + \frac{(x-x_1)^2}{2} f''(x_1) + \dots$

$\int_a^b f(x)\,dx = \int_{x_0}^{x_2} f(x)\,dx \approx \int_{x_0}^{x_2} \dots$ $\Rightarrow$

$$\Rightarrow \int_a^b f(x)\,dx \approx \frac{h}{3}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right], \quad h = \frac{b-a}{2}$$

Formulele compuse → vezi lab

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Integrale duble

$f: [a,b] \times [c,d] \to \mathbb{R}$

$$I = \int_a^b \int_c^d f(x,y)\,dx\,dy = ?$$

$$I = \int_a^b \left(\underbrace{\int_c^d f(x,y)\,dy}_{I_1} \right) dx$$

$$I_1 = \int_c^d f(x,y)\,dy \overset{\text{trapez simpler}}{\approx} \frac{d-c}{2}\left[f(x,c)+f(x,d)\right]$$

$$I = \int_a^b \frac{d-c}{2}\left[f(x,c)+f(x,d)\right]dx \overset{\text{trapez simpler}}{\approx}$$

$$\approx \frac{d-c}{2}\cdot\frac{b-a}{2}\left[f(a,c)+f(b,c)+f(a,d)+f(b,d)\right]$$

Metoda Romberg

$$\int_a^b f(x)\,dx = ?$$

$$\begin{bmatrix} R_{11} & & 0 \\ R_{21} & R_{22} & \\ \cdots & & \\ R_{n1} & R_{n2} & \cdots R_{nn} \end{bmatrix}$$

R_{k1}, $k = \overline{1:n}$, se calculează folosind formula compusă a trapezelor pentru 2^{k-1} subintervale

$$R_{11} = \frac{b-a}{2}\left[f(a)+f(b)\right]$$

$$R_{21} = \frac{b-a}{2}\left[f(a)+2f\left(\frac{a+b}{2}\right)+f(b)\right]$$

$$R_{ij} = R_{k,j-1} + \frac{R_{k,j-1} - R_{k-1,j-1}}{4^{k-1}}$$

Integrare numerică - Cuadrateri Gaussiene

$\int_a^b f(x)\,w(x)\,dx = ?$

$$\boxed{\int_a^b f(x)\,w(x)\,dx \simeq \sum_{k=0}^{n} q_k f(x_k)}$$, unde

x_k sunt rădăcinile unui polinom ortogonal Q_k

q_k se calculează din condiția ca formula de integrare să fie exactă dacă $f(x) = 1, x, \dots, x^n$

Produsul scalar a două polinoame

$$\boxed{\langle P_i, P_j \rangle = \int_a^b P_i(x) \cdot P_j(x)\,w(x)\,dx}$$

Două polinoame sunt ortogonale dacă au produsul scalar zero. O

Polinoame ortogonale

	a	b	$w(x)$	Relație de recurență
Cebâșev de ordin 1	-1	1	$\frac{1}{\sqrt{1-x^2}}$	$T_{n+1} - 2xT_n + T_{n-1} = 0$ $T_0 = 1,\ T_1 = x$
Cebâșev de ordin 2	-1	1	$\sqrt{1-x^2}$	$T_{n+1} - 2x \cdot T_n + T_{n-1} = 0$ $T_0 = 1,\ T_1 = 2x$
Legendre	-1	1	1	$(n+1)L_{n+1} - (2n+1)xL_n + nL_{n-1} = 0$ $L_0 = 0,\ L_1 = x$
Laguerre	0	∞	e^{-x}	$G_{n+1} - (2n+1-x)G_n + n^2 G_{n-1} = 0$ $G_0 = 1,\ G_1 = 1-x$
Hermite	$-\infty$	∞	e^{-x^2}	$H_{n+1} - 2xH_n + 2nH_{n-1} = 0$ $H_0 = 1,\ H_2 = 2x$

Ex 1: $\int_{-1}^{1} f(x)dx \simeq q_0 f(x_0) + q_1 f(x_1) + q_2 f(x_2) + q_3 f(x_3)$

$q_0, q_1, q_2, q_3 = ?$

$x_0, x_1, x_2, x_3 = ?$

$w(x) = 1,\ a = -1,\ b = 1$

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Calculare x_i ;

Met. 1 : x_0, x_1, x_2, x_3 sunt răd. pol Lagendre de grad 4

$(n+1)L_{n+1} - (2n+1)xL_n + nL_{n-1} = 0$

$L_0 = 1, \quad L_1 = x$

$n = 1 \Rightarrow 2L_2 - 3xL_1 + L_0 = 0 \Rightarrow L_2 = \frac{3xL_1 - L_0}{2} = \frac{3x^2 - 1}{2}$

$n = 2 \Rightarrow L_3 = \frac{5x^3 - 3x}{2}$

$n = 3 \Rightarrow L_4 = \frac{35x^4 - 70x^2 + 3}{8}$

$L_4(x) = 0 \Rightarrow P = \left[\frac{35}{8} \quad 0 \quad -\frac{30}{8} \quad 0 \quad \frac{3}{8}\right]$

roots(P)

$x_0 = -0,86$

$x_1 = -0,34$

$x_2 = 0,34$

$x_3 = 0,86$

Met. 2

Fie polinomul $\Pi_4(x) = (x-x_0)(x-x_1)(x-x_2)(x-x_3) =$

$= x^4 + \alpha x^3 + \beta x^2 + \gamma x + \delta$

$\Pi_4(x)$ este ortogonal cu polinoame de grad < 4

$\int_{-1}^{1} \Pi_4(x) \cdot x^k dx = 0 \quad , \omega(x) = 1, \quad k = 0:3$

$\int_{-1}^{1} (x^4 + \alpha x^3 + \beta x^2 + \gamma x + \delta) x^k \, dx = 0 \quad , k = 0:3$

se rezolvă SEL $\Rightarrow \alpha = 0, \beta = -\frac{6}{7}, \gamma = 0, \delta = \frac{3}{35}$

$x^4 - \frac{6}{7}x^2 + \frac{3}{35} = 0 \Rightarrow x_i = \dots$

Calculeaza a_0, a_1, a_2, a_3

$f(x)=1 \Rightarrow \int_{-1}^{1} dx = a_0 + a_1 + a_2 + a_3$

$f(x)=x \Rightarrow \int_{-1}^{1} x\,dx = a_0 x_0 + a_1 x_1 + a_2 x_2 + a_3 x_3$

$f(x)=x^2 \Rightarrow \int_{-1}^{1} x^2 dx = a_0 x_0^2 + a_1 x_1^2 + a_2 x_2^2 + a_3 x_3^2$ $\Big| \cdot 2)$

$f(x)=x^3 \Rightarrow \int_{-1}^{1} x^3 dx = \dots$

$\xrightarrow{\text{rez}}$ SEL $\Rightarrow a_0 = a_3 = 0,65$

$a_1 = a_2 = 0,35$

Ex 2

$$\int_{-1}^{1} f(x)\,dx \cong a_0 f(-1) + a_1 f(x_1) + a_2 f(x_2) + a_3 f(1)$$

$\omega(x) = 1$

Calculeaza x_1, x_2

Fie polinomul $\Pi_4(x) = (x+1)(x-x_1)(x-x_2)(x-1) =$

$= (x^2-1)(x^2 - Sx + P)$

unde $S = x_1 + x_2$

$P = x_1 \cdot x_2$

$\Pi_4(x)$ este ortogonal cu orice polinom de grad <4

$\int_{-1}^{1} \Pi_4(x) \cdot x^k dx = 0, \; k = 0 \div 3 \Rightarrow$ se rezolva SEL $\Rightarrow$

$\Rightarrow S = 0, \; P = -\frac{1}{3} \Rightarrow x^2 - \frac{1}{3} = 0 \Rightarrow x_1 = -\frac{\sqrt{3}}{3}, \; x_2 = \frac{\sqrt{3}}{3}$

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Calculare a_1, a_2, a_3, a_4

Daca $f(x) = 1, x, x^2, x^3$ =>

=> $\int_{-1}^{1} f(x)dx = a_0 + a_1 + a_2 + a_3$

$\int_{-1}^{1} x dx = \sum_{i=0}^{3} a_i x_i$

$\int_{-1}^{1} x^2 dx = \sum_{i=0}^{3} a_i x_i^2$

$\int_{-1}^{1} x^3 dx = \sum_{i=0}^{3} a_i x_i^3$

=> SEL => $a_i = \dots$

Ex 3 $\int_{-1}^{1} \frac{f(x)\,dx}{\sqrt{1-x^2}} \approx a_0 f(x_0) + a_1 f(x_1) + a_2 f(x_2)$

$\omega(x) = \frac{1}{\sqrt{1-x^2}}$

x_0, x_1, x_2 sunt rad. pol. Cebaşev de grad 3

$T_3(x) = 4x^3 - 3x$

$x_0 = \frac{-\sqrt{3}}{3}, \quad x_1 = 0, \quad x_2 = \frac{\sqrt{3}}{2}$

$f(x) = 1 \Rightarrow \int_{-1}^{1} \frac{dx}{\sqrt{1-x^2}} = \sum_{i=0}^{3} a_i$

$f(x) = x \Rightarrow \int_{-1}^{1} \frac{x\,dx}{\sqrt{1-x^2}} = \sum_{i=0}^{3} a_i x_i$

$f(x) = x^2 \Rightarrow \int_{-1}^{1} \frac{x^2 dx}{\sqrt{1-x^2}} = \sum_{i=0}^{3} a_i x_i^2$

$\Rightarrow a_0 = a_1 = a_2 = \frac{\pi}{3}$

Ex 4:

$I = \int_0^1 \frac{x^4 dx}{\sqrt{x(1-x)}}$

4p

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$[0,1] \to [-1,1]$

$x = ut + v$

$\begin{cases} 0 = -u + v \\ 1 = u + v \end{cases} \Rightarrow u = v = \frac{1}{2}$

$x = \frac{t+1}{2}$

$dx = \frac{1}{2} dt$

$I = \frac{1}{2} \int_{-1}^{1} \frac{\left(\frac{t+1}{2}\right)^4}{\sqrt{\frac{t+1}{2}\left(1-\frac{t+1}{2}\right)}} dt = \int_{-1}^{1} \frac{\left(\frac{t+1}{2}\right)^4}{\sqrt{1-t^2}} dt$

Fie $f(t) = \left(\frac{t+1}{2}\right)^4 \Rightarrow I = \int_{-1}^{1} \frac{f(t)}{\sqrt{1-t^2}} dt$

$\Rightarrow$ exact ca ~~la~~ ex 3

Ex 5:

$\int_a^b f(x)dx \approx a_0 f(x_0) + a_1 f(x_1) + a_2 f(x_2)$

$[a,b] \to [-1,1]$

$x = ut + v$

$\begin{cases} a = -u + v \\ b = u + v \end{cases} \Rightarrow u = \frac{b-a}{2}, \; v = \frac{a+b}{2}$

$x = \frac{b-a}{2} t + \frac{a+b}{2}, \; dx = \frac{b-a}{2} dt$

$I = \int_a^b f(x)dx = \frac{b-a}{2} \int_{-1}^{1} f\left(\frac{b-a}{2} t + \frac{a+b}{2}\right) dt$

Not $g(t) = f\left(\frac{b-a}{2} t + \frac{a+b}{2}\right)$

$I = \frac{b-a}{2} \int_{-1}^{1} g(t)dt$

$I_1 = \int_{-1}^{1} g(t)dt \approx b_0 g(t_0) + b_1 g(t_1) + b_2 g(t_2)$

t_0, t_1, t_2 rād. pol. Legendre de gr 3.

2) $x_i = \frac{b-a}{2} t_i + \frac{a+b}{2}$, $i = \overline{0,2}$

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Ex 6:

$I = \int_{-\infty}^{a} e^x f(x)\,dx \approx a_0 f(x_0) + a_1 f(x_1) + a_2 f(x_2)$

$[-\infty, a] \to [-a, \infty]$

$x = -t$

$dx = -dt$

$I = -\int_{\infty}^{-a} e^{-t} f(-t)\,dt = \int_{-a}^{\infty} e^{-t} f(-t)\,dt$

$[-a, \infty] \to [0, \infty]$

$y = t + a \Rightarrow t = y - a$

$dy = dt$

$I = \int_0^{\infty} e^{a-y} f(-y+a)\,dy$

Fie $g(y) = f(-y+a)$

$I_1 = \int_0^{\infty} e^{-y} g(y)\,dy \approx b_0 g(y_0) + b_1 g(y_1) + b_2 g(y_2)$ (*)

y_0, y_1, y_2 rădăcinile polinomului Laguerre de grad 3

$G_{n+1} - (2n+1-y) G_n + n^2 G_{n-1} = 0, \; G_0 = 1, \; G_1 = 1 - y$

calculăm $\Rightarrow G_3 = \frac{1}{6}(5-y)(y^2 - 4y + 2) - 4(1-y)$

$G_3(y) = 0 \Rightarrow y_0, y_1, y_2$

$x_i = -y_i + a, \; i = \overline{0,2}$

Pt. a calcula b_0, b_1, b_2, punem cond. ca formula (*) să fie exactă dacă $g(y) = 1, y, y^2$

$\int_0^{\infty} e^{-y}\,dy = b_0 + b_1 + b_2$

$\int_0^{\infty} y e^{-y}\,dy = b_0 y_0 + b_1 y_1 + b_2 y_2$

$\int_0^{\infty} y^2 e^{-y}\,dy = \sum_{i=0}^{2} b_i y_i^2$

$\Rightarrow$ SEL $\Rightarrow b_0 = \dots$

$b_1 = \dots$

$b_2 = \dots$

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Fie $I = [a, b]$

$y : I \to \mathbb{R}$

$f : I \times \mathbb{R} \to \mathbb{R}$

Problema diferențială de ord 1:

$y'(t) = f(t, y(t))$, $y(t) = ?$

Problema Cauchy de ord 1:

$y'(t) = f(t, y(t))$, $y(t_0) = \lambda_0$, $t_0 = a$, $y(t) = ?$

Fie $t_i = a + i \cdot h$, $i = 0 : n$, $h = \frac{b-a}{n}$

$y(t_i) = ?$ $i = 0 : n$

Metoda Euler (met. Runge-Kutta de ord. 1)

$$y(t) = y(t_i) + (t - t_i) \cdot y'(t_i) + \frac{(t-t_i)^2}{2} y''(t_i) + \dots$$

Dacă $t = t_{i+1}$

$$y(t_{i+1}) = y(t_i) + (t_{i+1} - t_i) y'(t_i)$$

Not. $y_i = y(t_i)$

$$y_{i+1} = y_i + h \cdot f(t_i, y_i), \quad i = 0 : n$$
$$y_0 = \lambda_0$$

$$\begin{cases} k = h \cdot f(t_i, y_i) \\ y_{i+1} = y_i + k \end{cases}$$

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Metoda Runge-Kutta de ord. 2

$$k_1 = h \cdot f(t_i, y_i)$$
$$k_2 = h \cdot f(t_i + h, y_i + k_1)$$
$$y_{i+1} = y_i + \frac{k_1 + k_2}{2}$$

$i = 0 : n$

$h = \frac{b-a}{n}$

$y_0 = \lambda_0$

Metoda Runge-Kutta de ord 4:

$$k_1 = h \cdot f(t_i, y_i)$$
$$k_2 = h \cdot f(t_i + \frac{h}{2}, y_i + \frac{k_1}{2})$$
$$k_3 = h \cdot f(t_i + \frac{h}{2}, y_i + \frac{k_2}{2})$$
$$k_4 = h \cdot f(t_i + h, y_i + k_3)$$
$$y_{i+1} = y_i + \frac{k_1 + 2k_2 + 2k_3 + k_4}{6}$$

$i = 0 : n$

$h = \frac{b-a}{n}$

$y_0 = \lambda_0$

Sisteme de n ecuații diferențiale de ord 1

$$\begin{cases} y_1' = f_1(t, y_1, y_2, \dots, y_n), \ y_1(t_0) = \lambda_{10} \\ y_2' = f_2(t, y_1, y_2, \dots, y_n), \ y_2(t_0) = \lambda_{20} \\ \vdots \\ y_n' = f_n(t, y_1, y_2, \dots, y_n), \ y_n(t_0) = \lambda_{n0} \end{cases}$$

$y_i(t)$ sec. ; $y_1(t_i), y_2(t_i), \dots, y_n(t_i) = ?$ 44 42, 57

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Dacă $n=2$:

$$\begin{cases} y_1' = f_1(t, y_1, y_2), \; y_1(t_0) = \lambda_{10} \\ y_2' = f_2(t, y_1, y_2), \; y_2(t_0) = \lambda_{20} \end{cases}$$

$y_1(t_i), y_2(t_i) = ?$ $i = 0:2$

$$k_{11} = h \cdot f_1(t_i, y_1^{(i)}, y_2^{(i)})$$

$$k_{12} = h \cdot f_2(t_i, y_1^{(i)}, y_2^{(i)})$$

$$k_{21} = h \cdot f_1\left(t_i + \frac{h}{2}, y_1^{(i)} + \frac{k_{11}}{2}, y_2^{(i)} + \frac{k_{12}}{2}\right)$$

$$k_{22} = h \cdot f_2\left(t_i + \frac{h}{2}, y_1^{(i)} + \frac{k_{11}}{2}, y_2^{(i)} + \frac{k_{12}}{2}\right)$$

$$k_{31} = h \cdot f_1\left(t_i + \frac{h}{2}, y_1^{(i)} + \frac{k_{21}}{2}, y_2^{(i)} + \frac{k_{22}}{2}\right)$$

$$k_{32} = h \cdot f_2\left(t_i + \frac{h}{2}, y_1^{(i)} + \frac{k_{21}}{2}, y_2^{(i)} + \frac{k_{22}}{2}\right)$$

$$k_{41} = h \cdot f_1(t_i + h, y_1^{(i)} + k_{31}, y_2^{(i)} + k_{32})$$

$$k_{42} = h \cdot f_2(t_i + h, y_1^{(i)} + k_{31}, y_2^{(i)} + k_{32})$$

$$y_1^{(i+1)} = y_1^{(i)} + \frac{k_{11} + 2k_{21} + 2k_{31} + k_{41}}{6}$$

$$y_2^{(i+1)} = y_2^{(i)} + \frac{k_{12} + 2k_{22} + 2k_{32} + k_{42}}{6}$$

$i = 0:n$, $h = \frac{b-a}{n}$, $y_1^{(0)} = \lambda_{10}$, $y_2^{(0)} = \lambda_{20}$

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Rezolvare ecuații diferențiale de ord. n

$y^{(n)}(t) = f(t, y(t), y'(t), \dots, y^{(n-1)}(t))$

$y(t_0) = \lambda_{10}$, $y'(t_0) = \lambda_{20}$, $\dots$, $y^{(n-1)}(t_0) = \lambda_{n0}$

$y(t)$ necunoscuta

$y(t_i) = ?$ $i = 0 : n$

Fie $u_1(t) = y(t)$

$u_2(t) = y'(t)$

$u_3(t) = y''(t)$

$\vdots$

$u_n(t) = y^{(n-1)}(t)$

$$\begin{cases} u_1'(t) = u_2(t), & u_1(t_0) = y(t_0) = \lambda_{10} \\ u_2'(t) = u_3(t), & u_2(t_0) = y'(t_0) = \lambda_{20} \\ \vdots \\ u_n'(t) = y^{(n)}(t) = f(t, u_1(t), u_2(t), \dots, u_n(t)) \\ u_n(t_0) = y^{(n-1)}(t_0) = \lambda_{n0} \end{cases}$$

Sistem de n ecuații diferențiale de ord. 1 cu necunoscute $u_1(t), u_2(t), \dots, u_n(t)$

$y(t_i) = u_1(t_i)$

$i = 0 : n$

2

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Ex: $t^2y'' + ty' - 4y = -3t, \quad t \in [1, 3]$

$y(1) = 4$

$y'(1) = 3$

$$y'' = -\frac{3}{t} - \frac{1}{t}y' + \frac{4}{t^2}y$$

$$y'' = f(t, y, y')$$

Fie $u_1(t) = y(t)$

$u_2(t) = y'(t)$

$$\begin{cases} u_1'(t) = u_2(t), & u_1(1) = y(1) = 4 \\ u_2'(t) = y''(t) = -\frac{3}{t} - \frac{1}{t}u_2(t) + \frac{4}{t^2}u_1(t), & u_2(1) = y'(1) = 3 \end{cases}$$

$u_1'(t) = f_1(t, u_1(t), u_2(t))$

$u_2'(t) = f_2(t, u_1(t), u_2(t))$

$f_1 = @(t, u_1, u_2) \; u_2$

$f_2 = @(t, u_1, u_2) \; -\frac{3}{t} - \frac{1}{t}u_2 + \frac{4}{t_2}u_1$

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Metode numerice cu pași legați

$y'(t) = f(t, y(t))$

$y(t_0) = \lambda_0$

$t_i = a + ih$, $i = 0 : n$, $h = \frac{b-a}{n}$

$y(t)$ nec.

$y(t_i) = ?$, $i = 0 : n$

Not: $y_i = y(t_i)$

$f_i = f(t_i, y_i)$

Metoda Adams-Bashforth

(1) $y_{i+1} = y_i + h \cdot \sum_{j=0}^{r-1} \beta_j f_{i-j}$, $i = 0 : n$

r s.n. ordinul metodei

Calculare β_j, $j = 0 : r-1$

- punctele $t_{i-r+1}, \dots, t_{i+1}$ se consideră echidistante cu $h = 1$ și $t_{i-r+1} = 0$
- se pune condiția ca formula (1) să fie exactă dacă $y(t) = 1, t, \dots, t^r$

dacă $r = 3$:

$$y_{i+1} = y_i + h(\beta_0 f_i + \beta_1 f_{i-1} + \beta_2 f_{i-2})$$

~~$y(t_{i+1}) = y(t_i)$~~

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Calculam $\beta_0, \beta_1, \beta_2$:

$h = 1,\ t_{i-2} = 0,\ t_{i-1} = 1,\ t_i = 2,\ t_{i+1} = 3$

$y(t) = 1 \Rightarrow f(t,y) = 0$

$1 = 1\ (A)$

$y(t) = t \Rightarrow f(t,y) = 1$

$t_{i+1} = t_i + (\beta_0 + \beta_1 + \beta_2) \Rightarrow \beta_0 + \beta_1 + \beta_2 = 1$

$y(t) = t^2 \Rightarrow f(t,y) = 2t$

$t_{i+1}^2 = t_i^2 + (\beta_0 \cdot 2t_i + \beta_1 \cdot 2t_{i-1} + \beta_2 \cdot t_{i-2})$

$9 = 4 + (4\beta_0 + 2\beta_1) \Rightarrow 4\beta_0 + 2\beta_1 = 5$

$y(t) = t^3 \Rightarrow f(t,y) = 3t^2$

$t_{i+1}^3 = t_i^3 + (\beta_0 \cdot 3t_i^2 + \beta_1 \cdot 3t_{i-1}^2 + \beta_2 \cdot 3t_{i-2}^2)$

$27 = 8 + (12\beta_0 + 3\beta_1)$

Se rezolvă SEL $\Rightarrow \beta_0 = \frac{23}{12},\ \beta_1 = -\frac{4}{3},\ \beta_2 = \frac{5}{12}$

$$y_{i+1} = y_i + \frac{h}{12}(23f_i - 16f_{i-1} + 5f_{i-2}),\ i = 2:n$$

$y_0 = \lambda$

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Metoda Adams - Moulton

$$y_{i+1} = y_i + h \cdot \sum_{j=-1}^{n-1} \beta_j f_{i-j}, \quad y_0 = \lambda \quad (2)$$

↑ ord. met.

Calculare β_j

- alegem punctele $t_{i-n+1}, \dots, t_{i+1}$ echidistante cu pasul $h=1$ și $t_{i-n+1} = 0$
- se pune condiția ca formula (2) să fie exactă dacă $y(t) = 1, t, \dots, t^{n+1}$

Dacă $n=3$:

$$y_{i+1} = y_i + h[\beta_{-1} f_{i+1} + \beta_0 f_i + \beta_1 f_{i-1} + \beta_2 f_{i-2}]$$

calculare $\beta_{-1}, \beta_0, \beta_1, \beta_2$:

$h=1, t_{i-2}=0, t_{i-1}=1, t_i=2, t_{i+1}=3$

$y(t)=1 \Rightarrow f(t, y(t)) = 0$

$l=1$ (A)

$y(t)=t \Rightarrow f(t, y(t)) = 1$

$t_{i+1} = t_i + (\beta_{-1} + \beta_0 + \beta_1 + \beta_2)$

$\Rightarrow \beta_{-1} + \beta_0 + \beta_1 + \beta_2 = 1$

$y(t)=t^2 \Rightarrow f(t, y(t)) = 2t$

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$t_{i+1}^2 = t_i^2 + (\beta_{-1} \cdot 2t_{i+1} + \beta_0 \cdot 2t_i + \beta_1 \cdot 2t_{i-1} + \beta_2 t_{i-2})$

$9 = 4 + (6\beta_{-1} + 4\beta_0 + 2\beta_1)$

$y(t) = t^3 \Rightarrow f(t, y(t)) = 3t^2$

$t_{i+1}^3 = t_i^3 + (\beta_{-1} \cdot 3t_{i+1}^2 + \beta_0 \cdot 3t_i^2 + \beta_1 \cdot 3t_{i-1}^2 + \beta_2 \cdot 3t_{i-2}^2)$

$2t = 8 + (\beta_{-1} \cdot 27 + \beta_0 \cdot 12 + \beta_1 \cdot 3)$

$y(t) = t^4 \Rightarrow f(t, y(t)) = 4t^3$

$t_{i+1}^4 = t_i^4 + (\beta_{-1} 4t_{i+1}^3 + \beta_0 \cdot 4t_i^3 + \beta_1 \cdot 4t_{i-1}^3 + \beta_2 \cdot 4t_{i-2}^3)$

$81 = 16 + (\beta_{-1} 108 + \beta_0 \cdot 32 + \beta_1 \cdot 4)$

Se rezolvă SEL $\Rightarrow \beta_{-1} = \frac{9}{24}, \beta_0 = \frac{19}{24}, \beta_1 = -\frac{5}{24}, \beta_0 = \frac{1}{24}$

$$y_{i+1} = y_i + \frac{h}{24}[9f_{i+1} + 19f_i - 5f_{i-1} + f_{i-2}]$$
$$i = 2:n$$
$$y_0 = \lambda$$

Metode predictor - corector

$$\begin{cases} y_{i+1}^{pred} = y_i + \frac{h}{12}(23f_i - 16f_{i-1} + 5f_{i-2}), & i = 2:n \\ y_{i+1} = y_i + \frac{h}{24}[9 \cdot f(t_{i+1}, y_{i+1}^{pred}) + 19f_i - 5f_{i-1} + f_{i-2}], & i = 2:n \\ y_{i+1} = y_{i+1}^{cor}, & i = 2:n \end{cases}$$

!

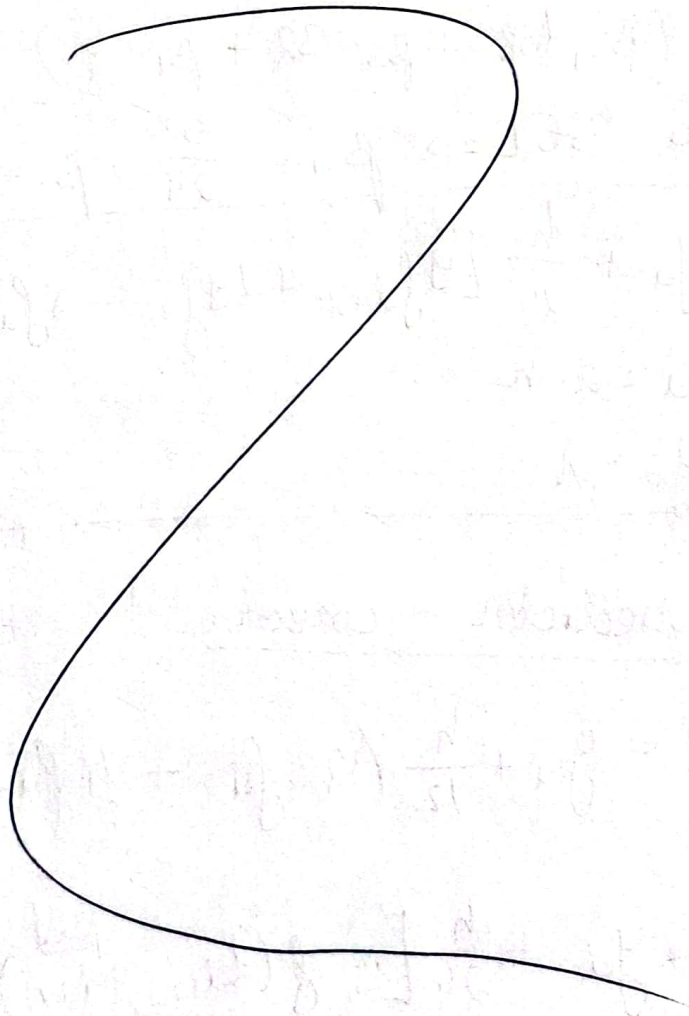
5!

$y = RK4(f, \lambda, q, 5, 4)$

$y = [y(t_0)\ y(t_1)\ y(t_2)\ y(t_3) \dots y(t_4)]$

$y = [y_0 \quad \underbrace{y_1\ y_2}_{RK4} \quad \underbrace{y_3 \dots y_4}_{\text{se calc. fol. met. pred - corecto}}]$

λ



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Aproximare numerică

Un spațiu prehilbertian este $(\mathcal{F}, u)$, unde $\mathcal{F}$ este un spațiu vectorial și u este produsul scalar

$$u : \mathcal{F} \times \mathcal{F} \to \mathbb{R}$$

$$\langle f, g \rangle = \int_a^b f(x) \cdot g(x) \cdot w(x)\, dx \quad , f, g \in \mathcal{F}$$

Proprietăți ale produsului scalar:

1) $\langle f, f \rangle \geq 0$

2) $\langle f_1 + f_2, f_3 \rangle = \langle f_1, f_3 \rangle + \langle f_2, f_3 \rangle$

3) $\langle f_1, f_2 \rangle = \langle f_2, f_1 \rangle$

4) $\langle c f_1, f_2 \rangle = c \cdot \langle f_1, f_2 \rangle$

Fie G un subspațiu finit al lui $\mathcal{F}$.
Fie elem. $f \in \mathcal{F}$. Cel mai bun aproximant în sensul c.m.m.p. pt. elem. $f \in \mathcal{F}$ în subsp. G este elementul $g^* \in G$ a.î.:

$$\|f - g\|_2 = \min_{g \in G} \{ \|f - g\|_2 \}$$

condiția necesară și suficientă ca g^* să fie cel mai bun aproximant în sensul c.m.m.p. pentru f este:

$$\boxed{\langle f \cdot g^*, g \rangle = 0}$$

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Fie o bază $\{u_0, u_1, \dots, u_n\}$ din G

$$g^* = \sum_{k=0}^{n} c_k^* \cdot u_k \quad , c_k^* = ?$$

Fie $g \neq g^* \in G$: $g = \sum_{j=0}^{n} c_j^* u_j$

$$\langle f - g^*, g \rangle = 0$$

$$\langle f - g^*, \sum_{j=0}^{n} c_j^* \cdot u_j \rangle = 0$$

$$\langle \sum_{j=0}^{n} c_j^* \cdot u_j, f - g^* \rangle = 0$$

$$\sum_{j=0}^{n} c_j^* \langle u_j, f - g^* \rangle = 0$$

$$\langle u_j, f - g^* \rangle = 0$$

$$\langle u_j, f - \sum_{k=0}^{n} c_k^* u_k \rangle = 0$$

$$\langle u_j, \sum_{k=0}^{n} c_k^* u_k \rangle = \langle u_j, f \rangle, \quad j = \overline{0:n}$$

$$\sum_{k=0}^{n} c_k^* \langle u_j, c_k^* u_k \rangle = \langle u_j, f \rangle, \quad j = \overline{0:n}$$

$$\Rightarrow \begin{bmatrix} \langle u_0, u_0 \rangle & \langle u_0, u_1 \rangle & \dots & \langle u_0, u_n \rangle \\ \langle u_1, u_0 \rangle & \langle u_1, u_1 \rangle & \dots & \langle u_1, u_n \rangle \\ \vdots & \vdots & & \vdots \\ \langle u_n, u_0 \rangle & \langle u_n, u_1 \rangle & \dots & \langle u_n, u_n \rangle \end{bmatrix} \begin{bmatrix} c_0^* \\ c_1^* \\ \vdots \\ c_n^* \end{bmatrix} = \begin{bmatrix} \langle u_0, f \rangle \\ \langle u_1, f \rangle \\ \vdots \\ \langle u_n, f \rangle \end{bmatrix}$$

SEL $(n+1) \times (n+1)$ cu mat. sunătă

Dacă $\{u_0, \dots, u_n\}$ bază ortogonală $\Rightarrow$ $\boxed{c_k^* = \frac{\langle u_k, f \rangle}{\langle u_k, u_k \rangle}}$

Dacă $\{u_0, \dots, u_n\}$ bază ortonormată $\Rightarrow$ $\boxed{c_k^* = \langle u_k, f \rangle}$ ȘI

Aproximare continuă în sensul celor mai mici pătrate

Forma produsului scalar în cazul continuu:

$$\langle f, g \rangle = \int_a^b f(x) \cdot g(x) \cdot w(x)\, dx \quad , f, g \in \mathcal{F}$$

Baza polinomială

$$\{u_0, u_1, \dots, u_n\} = \{1, x, \dots, x^n\}$$

$$g^*(x) = c_n^* \cdot x^n + c_{n-1}^* \cdot x^{n-1} + \dots + c_0^*$$

$$c_j^* = ?$$

ex: vezi lab → se creează un SEL

Baza trigonometrică

$$\left\{\frac{1}{\sqrt{2}}, \cos x, \sin x, \cos 2x, \sin 2x, \dots, \cos nx, \sin nx\right\}$$

$$g^*(x) = \frac{1}{\sqrt{2}} a_0 + \sum_{p=1}^{n} (a_p \cos px + b_p \sin px)$$

$$a_0 = \frac{1}{\pi\sqrt{2}} \int_0^{2\pi} f(x)\, dx$$

$$a_p = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos px\, dx$$

$$b_p = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin px\, dx \quad , p = \overline{1:n}$$

Baza Cebaşev

$$\left\{\frac{1}{\sqrt{2}}, T_1(x), T_2(x), \dots, T_n(x)\right\}$$

Pol. Cebaşev de ord n: $\boxed{T_n(x) = \cos(n \cdot \arccos x)}$

$n \geq 0$

55

Notam $\theta = \arccos x$, $\theta \in [0, \pi]$

$T_n(x) = \cos(n\theta)$

$T_{n+1}(x) = \cos n\theta \cos\theta - \sin n\theta \sin\theta$

$T_{n-1}(x) = \cos n\theta \cos\theta + \sin n\theta \cdot \sin\theta$

$T_{n+1}(x) + T_{n-1}(x) = 2\underbrace{\cos n\theta}_{T_n(x)} \underbrace{\cos\theta}_{x}$

$$T_{n+1}(x) + T_{n-1}(x) = 2x T_n(x), \quad T_0(x) = 1, \quad T_1(x) = x$$

$$\int_{-1}^{1} \frac{T_q(x) \cdot T_n(x)}{\sqrt{1-x^2}}\, dx = \begin{cases} 0, & q \neq n \\ \frac{\pi}{2}, & q = n \neq 0 \\ \pi, & q = n = 0 \end{cases}$$

$$g^*(x) = \frac{1}{\sqrt{2}} + \sum_{p=1}^{n} a_p \cdot T_p(x)$$

$$a_0 = \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}}\, dx$$

$$a_p = \frac{2}{\pi} \int_{-1}^{1} \frac{T_p(x) f(x)}{\sqrt{1-x^2}}\, dx \qquad p = 1:n$$

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Aproximare discretă în sensul celor mai mici pătrate

Forma produsului scalar în cazul discret:

$$\langle f, g \rangle = \sum_{i=0}^{n} f(x_i) \cdot g(x_i) \cdot \omega(x_i)$$

$f, g \in \mathcal{F}$

$$\| f - g^* \|_2 = \min_{g \in G} \{ \| f - g \|_2 \}$$

$$\| f - g \|_2 = \sqrt{\langle f - g, f - g \rangle} = \sqrt{\sum_{i=0}^{n} [f(x_i) - g(x_i)]^2 \cdot \omega(x_i)}$$

Baza polinomială

$\{1, x, \dots, x^n\}$

$$g^*(x) = c_n^* x^n + c_{n-1}^* x^{n-1} + \dots + c_0^*$$

$c_i^* = ?$

ex: calc. pol. din aprox în sensul cmmp de gr 2 care aprox:

x	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1
$f(x)$	1	$\frac{1}{2}$	0	$\frac{1}{2}$	1

, $\omega(x) = 1$

$$g^*(x) = c_2^* x^2 + c_1^* x + c_0^*$$

$$\begin{bmatrix} \langle u_0, u_0 \rangle & \langle u_0, u_1 \rangle & \langle u_0, u_2 \rangle \\ \langle u_1, u_0 \rangle & \langle u_1, u_1 \rangle & \langle u_1, u_2 \rangle \\ \langle u_2, u_0 \rangle & \langle u_2, u_1 \rangle & \langle u_2, u_2 \rangle \end{bmatrix} \cdot \begin{bmatrix} c_0^* \\ c_1^* \\ c_2^* \end{bmatrix} = \begin{bmatrix} \langle f, u_0 \rangle \\ \langle f, u_1 \rangle \\ \langle f, u_2 \rangle \end{bmatrix}$$

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$u_0(x) = 1$
$u_1(x) = x$
$u_2(x) = x^2$

$x_0 = -1$ $\quad x_3 = \frac{1}{2}$
$x_1 = -\frac{1}{2}$ $\quad x_4 = 1$
$x_2 = 0$

$$\langle u_1, u_1 \rangle = \sum_{i=0}^{4} x_i y_i \cdot 1$$

$$\langle u_2, u_2 \rangle = \sum_{i=0}^{4} x_i^2 \cdot x_i^2 = \dots$$

$$\vdots$$

$$\langle f, u_0 \rangle = \sum_{i=0}^{4} f(x_i) \cdot 1 \cdot 1$$

$$\vdots$$

Baza trigonometrică

$\left\{ \frac{1}{\sqrt{2}}, \cos x, \sin x, \cos 2x, \sin 2x, \dots, \cos hx, \sin hx \right\}$

$$g^*(x) = \frac{1}{\sqrt{2}} a_0 + \sum_{i=1}^{n} (a_p \cos px + b_p \sin px)$$

pol. de ord n

$$\sum_{k=0}^{2n-1} u_q(x_k) \cdot u_r(x_k) = \begin{cases} 0, & q \neq r \\ 1, & q = r \end{cases}$$

unde $x_k = \frac{k\pi}{n}$, $k = 0 : 2n-1$

$$a_0 = \frac{1}{\sqrt{2}} \sum_{k=0}^{2n-1} f(x_k)$$

$$a_p = \sum_{k=0}^{2n-1} f(x_k) \cdot \cos px_k, \quad p = 1 : n$$

$$b_p = \sum_{k=0}^{2n-1} f(x_k) \cdot \sin px_k, \quad p = 1 : n$$

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Ex: Să se calc. pol. de aprox. trig. în sensul celor mai mici pătrate de ordinul 2 pt:

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$
$f(x)$	-1	0	1	2

$x_k = \frac{k\pi}{2}$, $k = 0:3$

$$g^*(x) = \frac{1}{\sqrt{2}} a_0 + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x$$

$$a_0 = \frac{1}{\sqrt{2}} \sum_{k=0}^{3} f(x_k)$$

$$a_1 = \sum_{k=0}^{3} f(x_k) \cdot \cos x_k$$

$$b_1 = \sum_{k=0}^{3} f(x_k) \cdot \sin x_k$$

$$a_2 = \sum_{k=0}^{3} f(x_k) \cdot \cos 2x_k$$

$$b_2 = \sum_{k=0}^{3} f(x_k) \cdot \sin 2x_k$$

Baza Cebâşev

$$\left\{ \frac{1}{\sqrt{2}}, T_1(x), T_2(x), \dots, T_4(x) \right\}$$

$$g^*(x) = \frac{1}{\sqrt{2}} a_0 + \sum_{p=1}^{4} a_p T_p(x)$$

$$\sum_{k=0}^{4} \delta_p(x_k) \cdot \delta_n(x_k) = \begin{cases} 0, & p \neq n \\ \frac{n+1}{2}, & p = n \end{cases}$$

unde x_k sunt rădăcinile polinomului Cebâşev de grad $n+1$: $x_k = \cos \frac{2k+1}{2n+2} \pi$, $k = 0:n$ 59

$$a_0 = \frac{1}{\sqrt{2}} \cdot \frac{2}{n+1} \cdot \sum_{k=0}^{n} f(x_k)$$

$$a_p = \frac{2}{n+1} \sum_{k=0}^{n} T_p(x_k) \cdot f(x_k)$$

Stabilitate numerică

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RK4 – sisteme → generalizare:

$$K_{1,k} = h \cdot f_k\left(t_i, y_1^{(i)}, y_2^{(i)}, \dots, y_n^{(i)}\right)$$

$$K_{2,k} = h \cdot f_k\left(t_i + \frac{h}{2}, y_1^{(i)} + \frac{k_{11}}{2}, y_2^{(i)} + \frac{k_{12}}{2}, \dots, y_n^{(i)} + \frac{k_{1n}}{2}\right)$$

$$K_{3,k} = h \cdot f_k\left(t_i + \frac{h}{2}, y_1^{(i)} + \frac{k_{21}}{2}, \dots y_n^{(i)} + \frac{k_{2n}}{2}\right)$$

$$K_{4,k} = h \cdot f_k\left(t_i + h, y_1^{(i)} + k_{31}, \dots, y_n^{(i)} + k_{3n}\right)$$

$$y_k^{(i+1)} = y_k^{(i)} + \frac{k_{1k} + 2k_{2k} + 2k_{3k} + k_{4k}}{6}$$

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Algoritmi

MPD: (cea mai mare val proprie)

```
function [lambda, y] = MPD (A, y0, tol, max_iter)
    y = y0;
    lambda = Inf;
    for step = 1 : maxiter
        z = A * y;
        y = z / norm(z);
        lambda = y' * A * y;
        if (norm (A*y - lambda * y) < tol)
            return;
        endif
    endfor
endfunction
```

MPD^{-1}: (cea mai mică val proprie) → aste e de fapt MPI nedeplasat...

```
function [lambda, y] = MPD_mod (A, y0, tol, maxiter)
    y = y0; A_inv = inv(A);
    for step = 1 : maxiter
        z = A_inv * y;
        y = z / norm(z);
        lambda = y' * A_inv * y;
        if (norm (A_inv * y - lambda * y) < tol)
            lambda = 1 / lambda;
            return; endif, endfor, endfunction
```

MPi: (cu deplasare)

```
function [lambda, y] = MPi (A, y0, tol, maxiter, mi)
    y = y0; [n n] = size(A);
    lambda = y' * A * y;
    for step = 1 : maxiter
        % (A - mi*I) z = y
        mat = A - mi * eye(n);
        mat = inv(mat);
        z = mat * y;
        y = z / norm(z);
        lambda = y' * A * y;
        if (norm(A*y - lambda*y) < tol)
            return;
        endif
    endfor
endfunction
```

$$
\begin{cases}
z = (A - \mu I_n)^{-1} y^{(k-1)} \\
y^{(k)} = z / \|z\| \\
\lambda^{(k)} = (y^{(k)})^T \cdot A \cdot y^{(k)}
\end{cases}
$$

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INTERP. VANDERMONDE:

```
function pol = vander(x, y)        % → vect. cu f mare
    n = length(x);
    A = zeros(n);
    A(1:n, 1) = 1;
    for i = 1:n
        for j = 1:n
            A(i,j) = x(i).^(j-1);
        endfor
    endfor
    pol = linsolve(A, y');  % pol = inv(A)*y;
endfunction
```

ALG. DE CASTELJAN:

→ calc. cerbele Bezier / pol. Bernstein

→ mat. cu coord. pol.

```
function cp = de_casteljan(P, t)
    n = size(P,1);  beta = P;
    for j = 2:n
        for i = 1:n-j+1
            beta(i,:) = (1-t).*beta(i,:)+t.*beta(i+1
        endfor
    endfor
    cp = beta(1,:);
endfunction
```

$t \in [0,1]$

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